



Mahindra University Hyderabad
École Centrale School of Engineering
Minor II Examinations,

Program: B. Tech. Branch: CM Year: II Semester: I
Subject: Algebra (MA 2106)

Date: 30/10/2025
Time Duration: 90 Minutes

Start Time: 02.00 PM
Max. Marks: 18

Instructions:

1. There are 5 questions, all of which are compulsory.
2. Justify your answer wherever required.
3. All Notations are standard.

1. Prove the following:

[6] M

- (a) $SL(2, \mathbb{R})$ is a normal subgroup of $GL(2, \mathbb{R})$. [2]
- (b) The map $f : (\mathbb{R}, +) \rightarrow (\mathbb{R}^+, *)$ defined by $f(x) = e^x$ is a group homomorphism from the additive group of real numbers to the multiplicative group of positive real numbers. [2]
- (c) If $\phi : G \rightarrow G'$ is a group homomorphism then ϕ maps the identity element of G to the identity element of G' . [2]

2. (a) Define kernel of a group homomorphism. [1] [2] M
(b) Define zero-divisors in a commutative ring. [1]

3. (a) Write down the permutation $(235648)(134892)$ in S_9 as the product of disjoint cycles. [1] [6] M
(b) Find the permutation α in S_7 such that $\alpha(264537) = (1732)$. [1]
(c) What is the order of the permutation $(3429)(531786)$ in S_9 ? [2]
(d) List down all the elements of A_4 . [2]

4. Give an explicit example of a ring and a subring whose multiplicative identities are distinct. [2] M

5. Let N be a normal subgroup of a group G such that $a^{-1}b^{-1}ab \in N$ for all $a, b \in G$. Show that the factor group G/N is abelian. [2] M
